Introduction to Calculations of Heat loss on piping

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1 Introduction

This excel sheet allows to calculate heat losses for a single pipe with flowing liquid. It cannot be utilized to calculate heat losses on a piping network nor for gases. It utilizes a Python module available in Excel environment. The exact usage of this packet is described in the sheet "User's Guide". There are two modes of calculation:

- Normal - it is used for any calculations that do not involve any solar radiation.
- Solar radiation - it is used only if the piping is exposed to solar radiation. This mode should not be used if piping is inside building.

Those methods are described in more details in the next section.

2 Theory behind calculations

Both modes are utilizing the same energy balance and methodology. There are special additional considerations when it comes down to solar radiation mode, which are later explained. Here are main assumptions that allows us to solve energy balance:

1. Steady-state operation (both mass and energy).
2. The change in kinetic and potential energy is negligible.
3. The flow is turbulent and pipes are full of liquid.
4. The fluid is incompressible.
5. The specific heat capacity is constant.
6. Fluid is loosing energy to surroundings.
7. No axial heat transfer.
8. Uniform heat transfer distribution in piping cross section.
9. Piping is horizontal and enclosed by air only.
10. Air around piping is at ambient pressure.
11. Fluid is gaining energy by solar radiation (solar mode only).

With above in mind, one can utilize "Thermal Resistance Network" methodology and write that the average heat loss rate through a pipe segment of length L and distance x from the inlet is:

$$\dot{Q}_{avg}(x) = \frac{(T_f(x) - T_a)_{avg}}{R_{fa}} = 2\pi LK\left(T_f(x) - T_a\right)_{avg} \tag{1}$$

The *avg* notation means that we should take average of the temperature difference on piping segment of length L. With the same assumptions we can write that energy balance at any point must satisfy:

$$A_t\dot{q} - \dot{Q} = \dot{m}CdT_f \tag{2}$$

where:

- $T_f(x)$ - temperature of liquid at distance x, $(\mathrm{F}^\circ,\ \mathrm{C}^\circ)$
- T_a - ambient temperature, $(\mathrm{F}^\circ,\ \mathrm{C}^\circ)$
- A_t - total piping area $(\mathrm{ft}^2,\ \mathrm{m}^2)$
- $\dot{q}$ - solar radiation flux $\left(\frac{\mathrm{BTU}}{\mathrm{ft}^2},\ \frac{\mathrm{W}}{\mathrm{m}^2}\right)$
- R_{fa} - thermal resistance of piping of length from fluid to air L, $\left(\frac{\mathrm{F}^\circ}{\mathrm{BTU}}, \frac{\mathrm{C}^\circ}{\mathrm{W}}\right)$
- K - combined thermal conductivity, $\left(\frac{\mathrm{BTU}}{\mathrm{ft\cdot h\cdot F^\circ}}, \frac{\mathrm{W}}{\mathrm{m\cdot K}}\right)$
- $\dot{\mathrm{m}}$ - mass flow rate, $\left(\frac{\mathrm{lb}}{\mathrm{h}}, \frac{\mathrm{kg}}{\mathrm{s}}\right)$
- C - specific heat capacity, $\left(\frac{\mathrm{BTU}}{\mathrm{lb\cdot R}}, \frac{\mathrm{kJ}}{\mathrm{kg\cdot K}}\right)$

and we define K as:

$$\frac{1}{K} = \frac{2N}{h_f D_i} + \frac{1}{k_{pipe}} ln\left(\frac{D_o}{D_i}\right) + \frac{1}{k_{ins}} ln\left(\frac{D_o + 2y}{D_o}\right) + \frac{2N}{h(D_o + 2y)} \tag{3}$$

where:

- N - length conversion $\left(eq.\ 12\frac{\mathrm{inch}}{\mathrm{ft}}\right)$
- y - insulation thickness, inch
- D_i - inner diameter, inch
- D_o - outer diameter, inch
- h_f - heat transfer coefficient from fluid to piping wall, $\left(\frac{\mathrm{BTU}}{\mathrm{ft^2\cdot h\cdot F^\circ}}, \frac{\mathrm{W}}{\mathrm{m^2\cdot K}}\right)$
- h - combined heat transfer coefficient from surface of a piping to air, $\left(\frac{\mathrm{BTU}}{\mathrm{ft^2\cdot h\cdot F^\circ}}, \frac{\mathrm{W}}{\mathrm{m^2\cdot K}}\right)$
- k_{pipe} - thermal conductivity of a piping, $\left(\frac{\mathrm{BTU}}{\mathrm{ft\cdot h\cdot F^\circ}}, \frac{\mathrm{W}}{\mathrm{m\cdot K}}\right)$
- k_{ins} - thermal conductivity of a insulation, $\left(\frac{\mathrm{BTU}}{\mathrm{ft\cdot h\cdot F^\circ}}, \frac{\mathrm{W}}{\mathrm{m\cdot K}}\right)$

2.1 General solution

The presented approach does not calculate fluid temperature readily. Instead, surface temperature at $x = 0$ length is found, then integration is carried out to find surface temperature profile with length, and finally it is translated back to fluid temperature.

Before presenting general solution, it is important to remember that K and h are inherently functions of surface temperature T_s, which will be stated explicitly from now on. Let us also define two properties that simplifies the solution form.

$$\lambda = \frac{\dot{m}C}{2\pi}, \quad \mathrm{F}(T_s) = \frac{D_t h(T_s)}{2NK(T_s)}$$

With those two properties above, it can be shown that, there exist a set of parametric curves satisfying both equations (1) and (2) of form:

$$\mathrm{F}(T_s)\Delta T_{sa} - \Delta T_{fa} = 0 \tag{4}$$

$$\int_{T_{s,0}}^{T_s} \frac{1}{K(T_s)} \left(\frac{1}{\Delta T_{sa}} + \frac{d\ln\mathrm{F}(T_s)}{dT_s} \right) \left(1 - \frac{\dot{q}}{h(T_s)\Delta T_{sa}} \right)^{-1} dT_s + \frac{x}{\lambda} = 0 \tag{5}$$

With $\Delta T_{ia} = (T_i - T_a)$. Because above integral does not have a closed form solution, it must be solved with different methods. This can be easily plotted in any graphical tool, but that approach does not provide automation in finding solution for any provided case. It can also be solved with any root finding algorithm, but for any given x the solution lies very close to asymptotic value of $-\infty$, and from personal experience, root-finding approaches tend to make a lot iterations. For that very reason, curve (5) is solved with minimization algorithms. Before stating it mathematically, let us rewrite integrand as $I(T_s)$, then we can construct so-called loss function of form:

$$\mathcal{L}(x, T_s) = \left(\int_{T_{s,0}}^{T_s} I(T_s) dT_s + \frac{x}{\lambda} \right)^2 \tag{6}$$

And the final solution is found by minimizing this function on specially constructed interval (denoted Y), which is breafly explained in chapter XX. Mathematically, the surface temperature at given length is found by solving:

$$T_s(x) = \underset{T \in \mathrm{Y}}{\operatorname{argmin}}\ \mathcal{L}(x, T) \tag{7}$$

This value is then put back to (4) to recover fluid temperature.

2.2 Heat Transfer coefficient of piping surface

Heat transfer coefficient of air film around piping surface is calculated as a sum of radiation and convection effects. This provides a straightforward way of calculating both effects separately. Radiation effect is calculated with Stefan-Boltzman law, whereas convection effect is calculated with Churchill and Chu correlation for horizontal piping. That is:

$$\begin{aligned} h(T_s) &= h_{rad}(T_s) + h_{conv}(T_s) \\ h_{rad}(T_s) &= \epsilon \cdot 1.71 \cdot 10^{-9} \cdot (T_s^2 + T_a^2) \cdot (T_s + T_a) \\ h_{conv}(T_s) &= 9,636.27 \cdot \frac{\mu(T_{fl}) \cdot C_p(T_{fl})}{D_t} \left(0.6 + 0.321 \cdot \mathrm{Ra}^{1/6} \right)^2 \end{aligned} \tag{8}$$

where the equation for convective heat transfer coefficient comes from Churchill and Chu correlation assuming $\mathrm{Pr} = 0.72$ for all range of temperatures. Here, T_{fl}, μ and C_p are film temperature, dynamic viscosity and specific heat capacity of air at insulation-air interface, respectively, and D_t is total diameter in inch of piping segment. All of below equations are defined for SI units only. Users can use both Imperial and SI for calculations, and internally the python module uses SI metric for equations (10), (11), and (12).

Film temperature is just a mean between surface temperature and ambient temperature:

$$T_{fl} = \frac{T_s + T_a}{2} \, [\mathrm{K}] \tag{9}$$

We calculate dynamic viscosity from Sutherland's law for air:

$$\mu(T) = \frac{1.458 \cdot 10^{-6} \cdot T^{1.5}}{T + 110.4} [\mathrm{Pa} \cdot \mathrm{s}] \tag{10}$$

Polynomial relation for specific heat of air is taken from Cengel, Y. A., Boles, M. A., & Kanoglu, M. (2015). "Thermodynamics: An engineering approach" (8th ed.):

$$C_p(T) = 0.9703 + 6.7898 \cdot 10^{-5} \cdot T + 1.6576 \cdot 10^{-7} \cdot T^2 - 6.7863 \cdot 10^{-11} \cdot T^3 \left[\frac{\mathrm{kJ}}{\mathrm{kg} \cdot \mathrm{K}} \right] \tag{11}$$

Above equations (11) and (12) are valid in a range from around 200K to 1800K. There is no limit on upper or lower temperature in the spreadsheet, but below 273K specific heat of air is assumed to be equal to $C_p(273K)$ and above 1800K to be equal to $C_p(1800K)$.

In eq (8) Ra is the Rayleigh number which for air at low reduced pressure (below 0.1) can be calculated from:

$$\mathrm{Ra} = \frac{g\beta(T_s - T_a)D_t^3}{\nu\alpha} = 14.43 \cdot \frac{T_s - T_a}{T_{fl}^3 \cdot \mu(T_{fl})^2} D_t^3 \tag{12}$$

Here, g, β and α is the gravitational acceleration, isobaric thermal expansion coefficient and thermal diffusitivty of air, repspectively. This exact equation is obtained by assuming ambient pressure of 1 atm.

2.3 Minimization interval Y

Under assumptions mentioned in chapter 2, there are three cases that can be considered when solving (7):

1. $\dot{q} = 0$
2. $0 < \dot{q} < h(T_{s,0})\Delta T_{sa,0}$
3. $\dot{q} > h(T_{s,0})\Delta T_{sa,0}$

Here, (1) is simple to understand and Y is simply the bounds one would naturally choose:

$$\mathrm{Y} = [T_a, T_{s,0}] \tag{13}$$

In case of (2), the surface temperature is still decreasing with increasing length, but the lower bound is no longer ambient temperature due to solar incident radiation:

$$\mathrm{Y} = [T, T_{s,0}],\ \text{where } T \text{ s.t. } h(T)(T - T_a) = \dot{q} \tag{14}$$

In the last case (3), surface temperature increases with increasing length - fluid is heated by solar radiation. The lower bound is an ambient temperature, but upper bound is no longer initial surface temperature:

$$\mathrm{Y} = [T_a, T],\ \text{where } T \text{ s.t. } h(T)(T - T_a) = \dot{q} \tag{15}$$

It is important to note, that in case of fluid gaining heat from surroundings (inlet fluid temperature is lower than ambient temperature), there are only two cases to consider, either (1) or (3), bacause $\Delta T_{sa,0}$ will be negative or equal to 0, and Y will be (13) or (15).

3 Credits and References